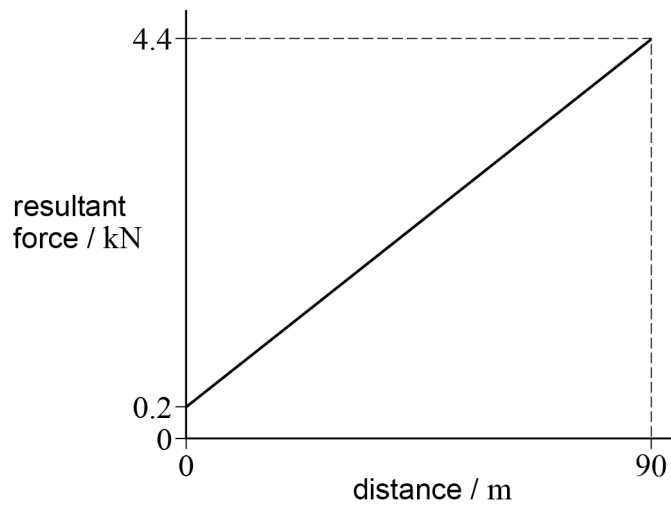


A car of mass 1400 kg starts from rest and travels along a horizontal road for 90 m
Figure 10 shows the variation with distance of the resultant force on the car.

Figure 10



0	9	.	1
---	---	---	---

Describe the motion of the car over the 90 m journey.

[1 mark]

0	9	.	2
---	---	---	---

Calculate the initial acceleration of the car.

[2 marks]

initial acceleration = _____ m s^{-2}



0 9 . 3

Show that the useful work done accelerating the car over the 90 m journey is about $2 \times 10^5 \text{ J}$

[3 marks]

0 9 . 4

Calculate the speed of the car after it has travelled 90 m

[2 marks]

final speed = _____ m s^{-2}

0 9 . 5

Discuss how the resistive forces experienced by the car change over the 90 m journey.

[2 marks]



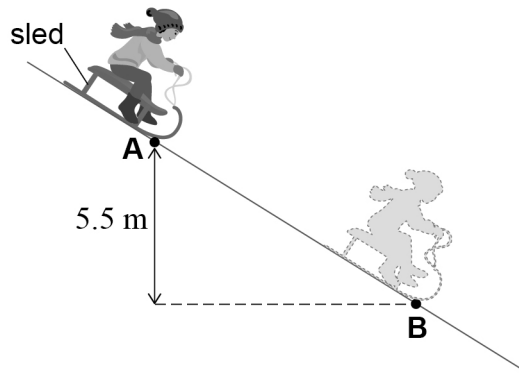
Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

Figure 5 shows a child on a sled sliding down a slope. The total mass of the child and the sled is 75 kg. She starts from rest at **A** and descends through a vertical height of 5.5 m to **B**.

Figure 5



State how to calculate the work done by a force.

[2 marks]

Show that the speed of the sled at **B** would be approximately 10 m s^{-1} in the absence of friction.

[2 marks]



0 9 . 3

The actual speed of the sled at **B** is 5.8 m s^{-1} .
The distance **AB** is 17.2 m.

Calculate using energy considerations the average frictional force experienced by the sled.

[4 marks]

average frictional force = _____ N

8

END OF SECTION A



Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

An apple falls from a tree.

What is the best estimate of its kinetic energy just before it hits the ground?

[1 mark]

A 30 mJ

☐

B 300 mJ

☐

C 3 J

☐

D 30 J

☐

Turn over for the next question

Turn over ►



Don't write	Don't write
-------------	-------------

2

1

A conveyor belt raises 20 boxes, each of mass 60 kg, through a vertical height of 12 m every 30 s. The conveyor belt operates at a steady speed.

What is the useful output power of the conveyor belt?

[1 mark]

A 0.47 kW

☐

B 4.7 kW

☐

C 14 kW

☐

D 140 kW

☐

2**2**

A lamp has a useful output power of 8.0 W and an efficiency of 12%.

What is the input power to the lamp?

[1 mark]

A 0.67 W

☐

B 0.96 W

☐

C 67 W

☐

D 96 W

☐

Determine the unit of work done expressed in fundamental (base) units.

[2 marks]

base units of work done = _____



Do not write

1 3 Which is a unit of power?

[1 mark]

A $\text{kg m}^2 \text{s}^{-3}$

☐

B $\text{kg m}^2 \text{s}^{-2}$

☐

C N m s^{-2}

☐

D $\text{N m}^{-1} \text{s}^{-1}$

☐

Turn over for the next question

Turn over ►



Do not write

Which row shows a vector quantity and a scalar quantity?

[1 mark]

	Vector quantity	Scalar quantity	
A	distance	kinetic energy	☐
B	force	displacement	☐
C	temperature	charge	☐
D	momentum	gravitational potential energy	☐

2**0**

An electric motor is used to lift a 120 g mass at a steady speed through a height of 0.15 m. The efficiency of the motor is 72%.

How much energy is transferred to the motor?

[1 mark]

A 0.05 J

☐

B 0.13 J

☐

C 0.25 J

☐

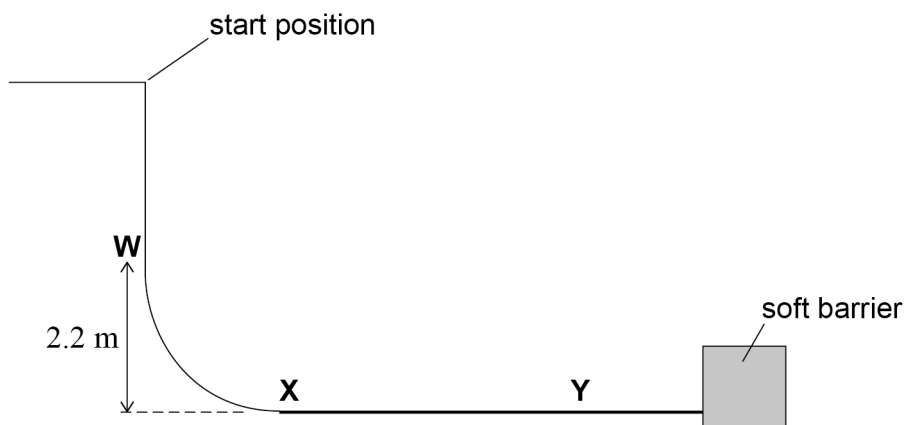
D 0.63 J

☐

0 6

Figure 4 shows a drop slide in a theme park.

Figure 4



W is the start of the curved section. **X** is the start of the horizontal section. A child travels along the horizontal section of the slide before coming to rest at **Y**. There is a soft barrier as a safety precaution.

Figures 5a, 5b and 5c represent the motion of the child on the vertical section, the curved section and the horizontal section of the drop slide.

Figure 5a

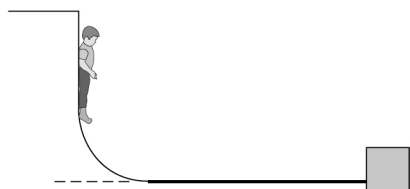


Figure 5b

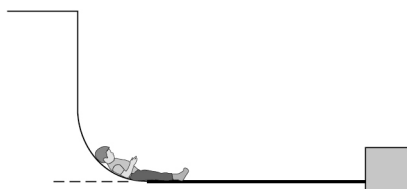
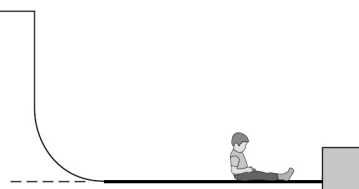


Figure 5c



0 6

1

A child of mass 40 kg using the drop slide has a speed of 7.5 m s^{-1} at **W**.

Show that the kinetic energy of the child at **X** is approximately 2 kJ.
Ignore friction between **W** and **X**.

[3 marks]



0 6 . 2 The average frictional force that acts on the moving child between **X** and **Y** is 250 N.

Calculate the distance **XY**.

[2 marks]

XY = _____ m

0 6 . 3 A man of mass 90 kg uses the slide. His speed as he hits the soft barrier is 3.2 m s^{-1} .
The horizontal section is 12.5 m in length.
He loses 90% of his kinetic energy along the horizontal section.

Calculate the average frictional force that acts on the man along the horizontal section.

[3 marks]

average frictional force = _____ N

0 6 . 4 Explain why a soft barrier is used to reduce the risk of injury rather than a rigid barrier.
[4 marks]



Do not write Do not write

1**1**

What is the unit of energy in SI (fundamental) base units?

[1 mark]

A N m

☐

B kg m s⁻¹

☐

C kg m² s⁻²

☐

D N m s⁻¹

☐

1**2**

Which row shows a scalar quantity and a vector quantity?

[1 mark]

	Scalar	Vector	
A	displacement	acceleration	☐
B	kinetic energy	displacement	☐
C	weight	speed	☐
D	distance	temperature	☐



Don't write

A box is pushed a distance x with a force F in a time t . The rate of energy transfer is P .

The same box is pushed a distance $2x$ with a force $5F$ in a time $\frac{t}{2}$.

What is the new rate of energy transfer?

[1 mark]

A $2P$

☐

B $5P$

☐

C $10P$

☐

D $20P$

☐

Don't write

An object is accelerated by a resultant force F and travels a distance s in the direction of the force. The average power used during the acceleration is P . The time taken is t .

A second object is accelerated using a resultant force of $3F$ and travels a distance $2s$ in the direction of the force. The average power developed during this acceleration is $0.5P$.

The second object accelerates for a time of

[1 mark]

A $0.33t$.

☐

B $1.3t$.

☐

C $3.0t$.

☐

D $12t$.

☐

Don't write

0 4

A train of mass 1.5×10^5 kg is travelling at a constant velocity on a horizontal track. The driving force F of the train is equal in magnitude to the resistive force R on the train.

0 4 . 1

The kinetic energy of the train is 5.5×10^7 J.
The train's engines produce a useful power of 3.7×10^6 W.

Calculate R .

[3 marks]

$R =$ _____ N

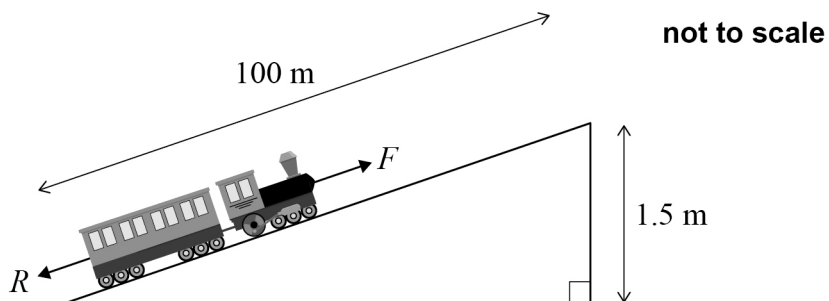
Question 4 continues on the next page



0 4 . 2

The train now ascends a slope of constant gradient. The train rises 1.5 m vertically for every 100 m of track.
 F is still equal in magnitude to R .

Figure 2



Calculate the acceleration of the train.

[3 marks]

acceleration = _____ m s^{-2}

6



Don't write

An LED lamp has a total power input of 7.0 W and an efficiency of 75%.

A filament lamp has a total power input of 65 W.

The useful power transferred by the filament lamp is double the useful power transferred by the LED.

Calculate the efficiency of the filament lamp.

[2 marks]

efficiency = _____



Don't write

1

1

An object of mass m falls from rest through a vertical distance s , reaching a velocity v .

The work done by air resistance during this fall is:

[1 mark]

A $m\left(gs - \frac{v^2}{2}\right)$ ☐

B $mgs + \frac{v^2}{2}$ ☐

C $m\left(\frac{v^2}{2} - gs\right)$ ☐

D $mgs - \frac{v^2}{2}$ ☐



Do not write

1

9

A car is being driven at a speed of 13.8 m s^{-1} .

The driver takes a time of 0.72 s to react to a hazard and apply the brakes.

The brakes cause a deceleration of 6.9 m s^{-2} which brings the car to rest.

What is the total distance travelled by the car from the point at which the driver sees the hazard?

[1 mark]

A 10 m

☐

B 14 m

☐

C 24 m

☐

D 38 m

☐

2**0**

What is the unit of power in SI fundamental (base) units?

[1 mark]

A kg m s^{-1}

☐

B kg m s^{-2}

☐

C $\text{kg m}^2 \text{s}^{-2}$

☐

D $\text{kg m}^2 \text{s}^{-3}$

☐

2**1**

A car of mass 1200 kg is travelling up a slope with an incline of 15° .
The car travels at a constant velocity of 20 m s^{-1} .
The motor provides a constant power output of 150 kW.

What is the magnitude of the resistive forces acting on the car?

[1 mark]

A 3000 N

☐

B 3900 N

☐

C 4500 N

☐

D 11 000 N

☐

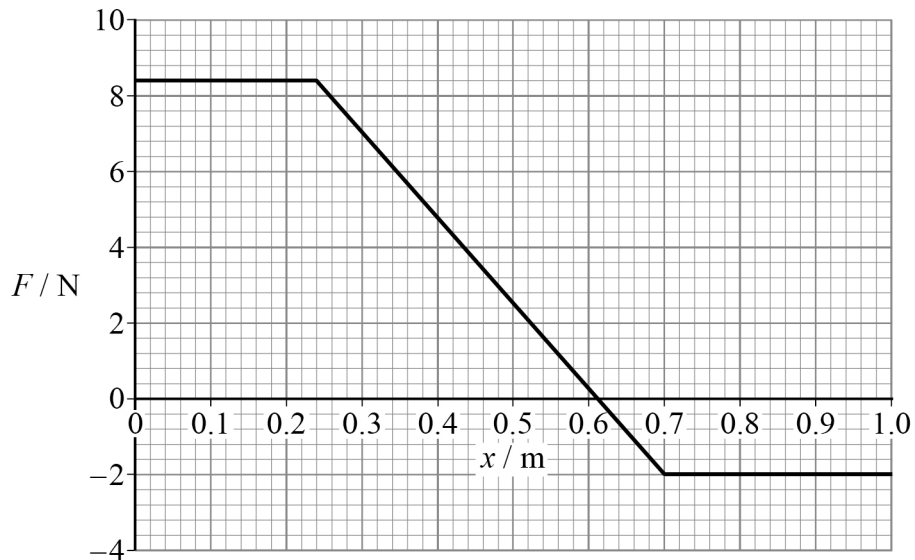
Do not write

An object is initially stationary at point **O**.

The object is acted on by a resultant force F and moves in a straight horizontal line away from **O**.

Figure 1 shows the variation of F with the displacement x of the object from **O**.

Figure 1



Determine the kinetic energy of the object when $x = 1.0$ m.

[3 marks]

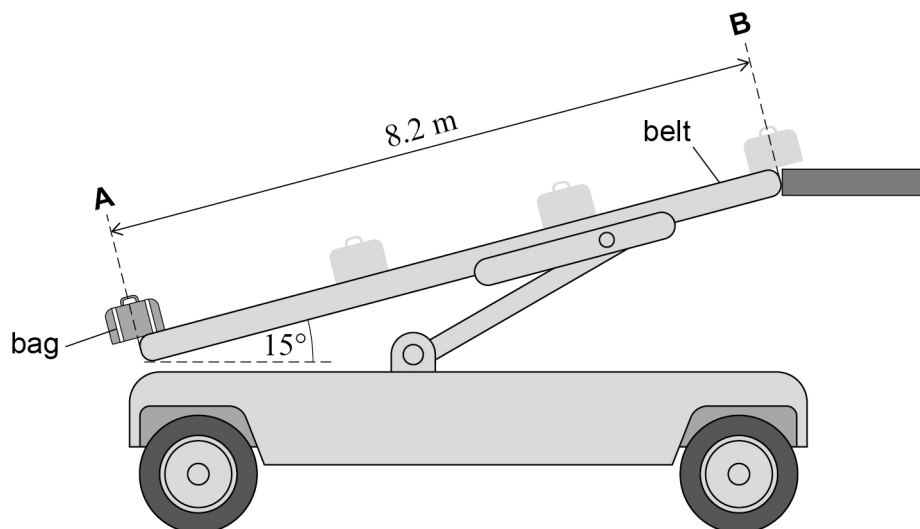
kinetic energy = _____ J

Don't write

Identical bags, each with a mass of 22 kg, are moved from point **A** to point **B** on a belt system as shown in **Figure 4**. Each bag travels a distance of 8.2 m along the belt and gains gravitational potential energy at the rate of 43 W.

Figure 4

not to scale



The belt is at an angle of 15° to the horizontal.

Calculate the time taken for a bag to move from **A** to **B**.

[3 marks]

time = _____ s



0 5 . 2

More bags are added to the belt without changing its speed.
The belt is driven by a motor that has an input power of 1.2 kW.
The maximum efficiency of the system is 31%.

Determine the maximum number of bags that can be on the belt at the same time
without changing the speed of the belt.

[2 marks]

maximum number of bags = _____

5

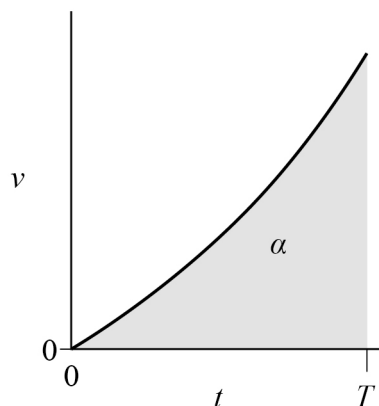
Turn over for the next question



Don't write

The thrust of a rocket engine has magnitude F and acts in the same direction as the direction of travel.

The graph shows the variation of the velocity v of the rocket with time t .



The area under the graph is α .

The gradient of the graph is μ at time $t = T$.

Which statement is correct?

[1 mark]

A At $t = T$, the acceleration of the rocket is $\frac{F}{\alpha}$

☐

B The total increase in kinetic energy of the rocket is $F\alpha$

☐

C The average mass of the rocket is $\frac{F}{\mu}$

☐

D At $t = T$, the power applied to the rocket is $F\mu$

☐


Don't write

A car is driven at a constant speed of 60 km hour^{-1} for 15 minutes. A constant resistive force of 180 N acts.

What is the work done against the resistive force?

[1 mark]

A 1.6 MJ

☐

B 2.7 MJ

☐

C 9.7 MJ

☐

D 11 MJ

☐

Don't write

2

4

A crane lifts a large steel bar of mass m at a constant speed v through a height h .
The crane's motor has an efficiency η .

What is the input power to the crane's motor?

[1 mark]

A $\frac{mgvh}{\eta}$

☐

B $\frac{mgv}{\eta}$

☐

C $mgvh\eta$

☐

D $mgv\eta$

☐



0	1
---	---

Determine the fundamental (base) unit of power.

[2 marks]

fundamental (base) unit of power = _____

— 2

2	0
---	---

A motorbike has an engine that is 27% efficient.
The motorbike does 14.6 MJ of useful work in a time of one hour.

What is the average input power?

[1 mark]

A 54 MW

☐

B 0.90 MW

☐

C 15 kW

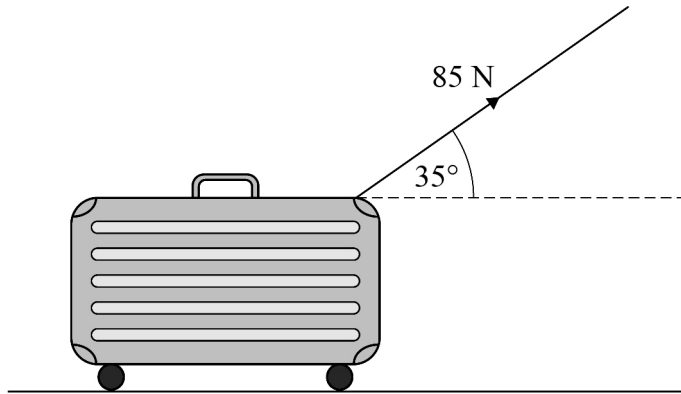
☐

D 4.1 kW

☐

2 2

A suitcase is pulled across a horizontal surface using a rope at an angle of 35° to the horizontal. The tension in the rope is 85 N. The suitcase is moved a distance of 100 m in a time of 1.5 minutes.



What is the average rate at which work is done?

[1 mark]

A 54 W ☐

B 77 W ☐

C 6010 W ☐

D 6900 W ☐

Turn over for the next question

Turn over ►



Don't write

0	2
---	---

An engineer tests a car by measuring the time taken for it to travel 1500 m at a constant speed. When delivering its maximum output power of 35 kW the time recorded is 28 s.

0	2
---	---

 .

1

Calculate the magnitude of the resistive force that is acting on the car.

[2 marks]

resistive force = _____ N

0	2
---	---

 .

2

Identify **one** way in which the engineer could modify the car so that the time to cover the test distance is reduced.

[1 mark]
